Physical Distributed GNSS Spoofing of UAV Swarms Evades Multi-Receiver Collapse Detection

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[TODO: affiliazione, email]

***Abstract*—Multi-receiver spoofing detection is regarded as one of the most robust countermeasures against civilian GNSS spoofing: receivers in a known formation compare the mutual distances implied by their GNSS fixes with the formation geometry, and a single-transmitter spoofer—which forces every victim onto the same position—is detected with near-certain probability. We evaluate this detector family against a *physical distributed* spoofer: 24 fixed ground transmitters deployed around the operating area of a 25-UAV swarm, each replaying the satellite closest to it in azimuth, with capture decided by received power through an elevation-dependent ground-to-UAV channel and with the replayed code phase propagating over the true three-dimensional path to every drone. No per-victim pre-compensation and no idealized attack are simulated. We implement the collapse detector of Jansen et al. (ACSAC 2016) in its three variants and a two-sided formation-consistency test representative of GNSS-vs-ranging checks, and evaluate them over 20 Monte Carlo missions per configuration with bootstrap confidence intervals. The physical attack displaces the swarm by 4 km while all three collapse detectors stay at $P_{\mathrm{D}} \leq 0.011$ (95 % upper bound) at every formation spacing from 5 to 100 m: a distributed spoofer does not collapse the formation, so the signature these detectors look for never appears. The consistency test retains only partial power, from $P_{\mathrm{D}} = 0.03$ at 5 m to 0.65 at 100 m at $P_{\mathrm{FA}} = 10^{-2}$, because the residual non-rigidity of ground transmitters (about 22 % of the baseline, caused by the mismatch between satellite and transmitter elevations) is masked by GNSS noise for compact swarms. We quantify this residual, show that it depends critically on deploying the array around the operating area, and delimit the result with respect to direction-of-arrival and absolute-reference defenses.**

***Index Terms*—GNSS spoofing, UAV swarms, spoofing detection, multi-receiver, distributed attack, relative distances.**

I. Introduction

Civilian GNSS signals are unauthenticated and can be forged by commodity software-defined radios [1]. Among the proposed countermeasures, *multi-receiver* detection stands out for its simplicity: since a single-antenna spoofer forces every victim in its coverage area to compute the same position fix [2], a group of receivers that knows its own formation can detect the attack by checking that the mutual distances implied by the GNSS fixes have not collapsed [3], [4]. The same principle underpins recent proposals for UAV swarms, which compare GNSS-derived inter-drone distances against independent ultra-wideband (UWB) ranging [5], [6].

This paper asks whether the guarantee survives a *distributed* attacker, i.e. several fixed ground transmitters surrounding the operating area. Our contributions are: (i) a physical simulation model of a distributed ground spoofer attacking a moving UAV swarm, with per-epoch satellite-to-transmitter routing, true three-dimensional propagation of the replayed signals, and an elevation-dependent ground-to-UAV channel [7]; (ii) a faithful implementation of the collapse detector of Jansen et al. [3] and of a two-sided formation consistency test [8], [5]; (iii) a Monte Carlo evaluation with confidence intervals showing that the physical attack defeats the collapse detectors entirely and the consistency test for compact swarms; and (iv) an analysis of the residual non-rigidity of ground transmitters and of its dependence on the array deployment, together with an honest delimitation of the result with respect to direction-of-arrival and absolute-reference defenses.

[TODO: aggiungere motivazione applicativa, struttura del paper]

II. Related Work

Multi-receiver position-domain detection: Tippenhauer et al. [2] formalized the GPS group-spoofing problem and proposed checking that the victims' computed positions preserve their physical formation; Jansen et al. [3] turned the check into a statistical test on pairwise distances with Rician/Rayleigh error models and spatially correlated noise, and Swaszek and Hartnett [4] derived Neyman–Pearson optimal variants for COTS receivers. Crowd-GPS-Sec [9] applies the same collapse principle to crowdsourced air-traffic receivers.

Measurement-domain and swarm variants: Stenberg et al. [8] test double differences of pseudorange and carrier phase across receivers (χ^2/GLRT), Kalantari and Larsson [10] derive a GLRT for receivers on a rigid body with unknown pose explicitly against multiple spoofers, and Borio and Gioia [11] detect the single common direction of arrival with a two-receiver sum-of-squares statistic. For UAV swarms, Mykytyn et al. [5] compare GNSS distances with UWB two-way ranging, and Bi et al. [6] cast detection as a semidefinite localization-feasibility problem. GNSS/INS fusion defenses such as SemperFi [12] and tightly coupled INS innovation monitors [13], [14] compare GNSS tracks with inertial dead reckoning.

Coordinated and distributed spoofing attacks: Result 7 of [2] proves that a formation-preserving group spoofing placement always exists (one transmitter per impersonated satellite, placed along the satellite direction); Jansen et al. [3] attempted a distance-preserving four-USRP attack and failed beyond one

receiver. Ceccato et al. [15] designed a multi-antenna ground spoofer with Wiener precoding that displaces a swarm without disrupting its formation; Chen et al. [16] analyze formation-preserving spoofing of small swarms; GNSS-WASP [17] demonstrated in hardware a transmitter constellation, placed in line with the satellites at altitude, that spoofs receivers over a wide area while preserving their relative distances; and Park and Yoo [18] state the translation-invariance of relative-only detectors as a formal proposition, rooted in the gauge freedom of network localization [19]. Surveys of the field include [20], [21], [22].

[TODO: posizionamento: a differenza di GNSS-WASP (trasmettitori in quota lungo le direzioni dei satelliti) e di Park–Yoo (traslazione ideale), qui l'attaccante e' un array fisico a terra e si misura il residuo osservabile]

III. System Model

A. Swarm, constellation, and measurement model

We consider a swarm of m UAVs flying in a square grid of $\sqrt{m} \times \sqrt{m}$ elements with spacing g (the *formation spacing*), following a way-point trajectory at altitudes between 100 and 700 m that spans about 7 km around its start point; positions are sampled every $T_s = 5$ s. The GNSS constellation is modeled as $N_{\mathrm{sat}} = 24$ satellites on 6 circular orbital planes (inclination $55°$, altitude 20 200 km) over a rotating spherical Earth. For UAV u at position $\mathbf{p}_u(t)$ and satellite s at position $\mathbf{s}_s(t)$ (visible when its elevation is positive), the authentic propagation delay is $\tau_{u,s}(t) = \|\mathbf{s}_s(t) - \mathbf{p}_u(t)\|/c$. Each receiver has a clock bias $\delta t_u \sim \mathcal{U}(-1, 1)$ s, constant over the run, and each pseudorange carries additive white Gaussian noise of standard deviation $\sigma_{\mathrm{pr}} = 3$ m.

Each UAV computes its position fix $\hat{\mathbf{p}}_u$ independently by iterated linearized least squares (Gauss–Newton) over the four unknowns $(x, y, z, \delta t)$: at iteration k, with predicted ranges $R_s = \|\mathbf{s}_s - \hat{\mathbf{x}}^{(k)}\|$ and pseudoranges $\rho_s = c\,(\tilde{\tau}_{u,s} - \widehat{\delta t}^{\,(k)})$,

$$\mathbf{G}\,\Delta = \boldsymbol{\rho} - \mathbf{R}, \qquad \mathbf{G}_{s,:} = \left[\frac{(\hat{\mathbf{x}} - \mathbf{s}_s)^{\top}}{R_s},\ c\right], \tag{1}$$

solved in the least-squares sense for five iterations, where $\tilde{\tau}_{u,s}$ is the *measured* delay (authentic or spoofed). We verified that, without attack and noise, the solver returns the true grid positions to numerical precision (error $< 10^{-4}$ m), so estimated and true positions live in the same frame.

In addition to GNSS, the UAVs measure their instantaneous mutual distances with an independent on-board ranging sensor (e.g. UWB two-way ranging [5]):

$$d_{ij}^{\mathrm{meas}}(t) = \|\mathbf{p}_i(t) - \mathbf{p}_j(t)\| + \varepsilon_{ij}(t), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_{\mathrm{rng}}^2), \tag{2}$$

with $\sigma_{\mathrm{rng}} = 0.1$ m. This is the most favorable setting for the defender: the true formation is known essentially without error at every epoch.

B. Ground-to-UAV channel

Spoofing signals travel over a ground-to-air link. We adopt the suburban cellular-to-UAV path-loss model of Al-Hourani and Gomez [7]: for a ground transmitter at horizontal distance d and elevation angle θ (in degrees) as seen from the UAV,

$$\mathrm{PL}(d, \theta) = 10\,\alpha \log_{10}(d) + A\,(\theta - \theta_0)\,e^{-\frac{\theta - \theta_0}{B}} + \eta_0 \quad [\mathrm{dB}], \tag{3}$$

with $\alpha = 3.04$, $A = -23.29$, $B = 4.14$, $\theta_0 = -3.61°$, $\eta_0 = 20.70$ dB (shadowing disabled for repeatability). The excess-loss term is fitted in [7] for $\theta \in [-2°, 10°]$; larger angles are saturated at $10°$. With the array deployment of Section IV no link exceeded this limit in any run (saturation fraction 0.000), so the model is always used within its validity range. Authentic signals follow free-space loss from satellite transmit power $P_t^{\mathrm{sat}} = 25$ W with antenna gains $G_t^{\mathrm{sat}} = 13.5$ dB and $G_r = 5$ dB, giving received powers of about -150 dBW.

IV. Physical Distributed Spoofing Attack

The attacker's goal is to displace the whole swarm's perceived position onto a decoy point (about 5.7 km from the flight path), while the swarm believes it is following its mission plan. The attack is simulated transmitter by transmitter; nothing is assumed about per-victim signal shaping.

A. Transmitter geometry and per-epoch routing

$N_{\mathrm{sp}} = 24$ fixed ground transmitters are placed on a circle of radius $R_{\mathrm{sp}} = 10$ km enclosing the swarm's operating area (Fig. 1), each radiating $P_t^{\mathrm{sp}} = 1$ mW with $G_t^{\mathrm{sp}} = 1$ dB. Following the geometric argument of [2, Result 7]—a formation-preserving group spoofer uses one transmitter per impersonated satellite, as close as possible to the satellite's direction—each visible satellite s is routed, at every epoch, to the transmitter whose azimuth as seen from the *current* true swarm centre $\mathbf{p}^{\mathrm{ctr}}(t)$ is closest to the satellite's azimuth:

$$\mathrm{sp}(s, t) = \arg\min_{k} \left|\angle_{2\pi}\big(\phi_s(t) - \phi_k(t)\big)\right|. \tag{4}$$

The transmitters never move; only the routing of satellites to transmitters is updated, which requires the attacker to track the swarm centre—the same knowledge it needs to calibrate the replayed delays.

B. Capture and replayed delay

For each UAV u and visible satellite s the receiver locks onto the transmitter whenever the received spoofing power, computed with (3), exceeds the authentic one. With the chosen EIRP the spoofer margin over the satellite is between 13 and 29 dB on every link (median 19.5 dB), so capture is complete and homogeneous across the swarm (no UAV mixes authentic and spoofed satellites).

Transmitter $k = \mathrm{sp}(s, t)$ replays the signal of satellite s with its code phase calibrated so that, at the true swarm centre, the measured delay equals the decoy delay τ_s^{dec} (the delay that satellite s would exhibit at the decoy position). A UAV at a different position receives the same transmission over a different path, so it measures

$$\tilde{\tau}_{u,s} = \tau_s^{\mathrm{dec}} - \delta t_u + \frac{\|\mathbf{q}_k - \mathbf{p}_u\| - \|\mathbf{q}_k - \mathbf{p}^{\mathrm{ctr}}\|}{c}, \tag{5}$$

where $\mathbf{q}_k$ is the transmitter position and both norms are true three-dimensional slant distances (altitude included). The

last term is the physical propagation residual: it is zero for the centre UAV, which therefore computes exactly the decoy position, and nonzero for the other UAVs.

C. Residual non-rigidity of a ground array

Write $\mathbf{p}_u = \mathbf{p}^{\mathrm{ctr}} + \boldsymbol{\delta}_u$ with $\|\boldsymbol{\delta}_u\| \ll \|\mathbf{q}_k - \mathbf{p}^{\mathrm{ctr}}\|$. To first order the residual in (5) is $-(\hat{\mathbf{w}}_k \cdot \boldsymbol{\delta}_u)/c$, where $\hat{\mathbf{w}}_k$ is the unit vector from the centre to the transmitter, while the delay that would make UAV u compute the decoy position *rigidly displaced* by $\boldsymbol{\delta}_u$ is $-(\hat{\mathbf{e}}_s \cdot \boldsymbol{\delta}_u)/c$, with $\hat{\mathbf{e}}_s$ the unit line of sight to the satellite. For a horizontal swarm and azimuth-matched routing the per-satellite pseudorange error is therefore

$$\epsilon_{u,s} \;\approx\; (\hat{\mathbf{a}}_s \cdot \boldsymbol{\delta}_u)\,\big(\cos\theta_k - \cos \mathrm{el}_s\big), \tag{6}$$

where $\hat{\mathbf{a}}_s$ is the satellite azimuth direction, θ_k the (near-zero) elevation of the ground transmitter and el_s the satellite elevation. A ground transmitter cannot reproduce the satellite's elevation, so (6) vanishes only for satellites near the horizon; it scales with the baseline and varies in sign across satellites, and the least-squares solver turns it into a position-dependent error, i.e. a *deformation* of the computed formation. Section VI measures this residual at about 22 % of the baseline. Two further facts, established numerically, are relevant for an attacker: the residual does not depend on transmitter power (capture is already complete at 1 mW), and it depends critically on the array enclosing the operating area—with a 2.5 km array centred on the start point the swarm is inside the circle only 26 % of the time, and the deformation grows from 12 % inside to 55 % outside (correlation 0.98 with the distance from the array centre), because from outside the array all transmitters lie in a narrow angular sector and no routing can match the satellite azimuths.

Remark 1 (Why collapse detectors are blind): The collapse detectors of Section V look for pairwise distances that *shrink* towards zero, the signature of a single transmitter seen by all receivers [2, Result 1]. A distributed spoofer with one transmitter per satellite delivers to every UAV a different set of delays, consistent to first order with its own displacement from the centre: the computed formation is translated and deformed by (6), but not collapsed. The statistic these detectors threshold therefore never enters the alarm region, independently of the threshold.

V. Relative-Distance Detection Methods

All detectors operate on the normalized distance ratios

$$r_{ij}(t) = \frac{\hat{d}_{ij}(t)}{d_{ij}^{\mathrm{meas}}(t)}, \qquad \hat{d}_{ij} = \|\hat{\mathbf{p}}_i - \hat{\mathbf{p}}_j\|, \tag{7}$$

computed over all $\binom{m}{2} = 300$ pairs at every epoch, where d_{ij}^{meas} is the instantaneous UWB measurement of (2); under authentic conditions $r_{ij} \approx 1$.

A. Collapse tests (Jansen et al.)

The ACSAC 2016 detector [3] declares spoofing when the pairwise distances *shrink*:

$$f\big(\{r_{ij}\}\big) \underset{H_1}{\overset{H_0}{\gtrless}} \lambda, \qquad \text{alarm if } f < \lambda, \tag{8}$$

with the three variants of its Appendix A implemented here:

$$f_{\min} = \min_{i<j} r_{ij}, \quad f_{\max} = \max_{i<j} r_{ij}, \quad f_{\mathrm{maj}} = \operatorname*{med}_{i<j} r_{ij}, \tag{9}$$

where the median form is equivalent to the paper's (normalized) majority voting: at least half of the pairs below λ iff the median is. Jansen et al. set λ from a Rician (H_0) / Rayleigh (H_1) model of the pairwise distance with spatially correlated position noise, $\sigma_d = \sqrt{2}\,\sigma\sqrt{1-\rho}$; we calibrate λ empirically to the same effect (Sec. V-C).

B. Formation consistency test

As a representative of the two-sided GNSS-vs-ranging consistency family — $|d_{\mathrm{GPS}} - d_{\mathrm{UWB}}| > d_{\mathrm{thr}}$ for UAV swarms [5], double-difference GLRTs across receivers [8], GLRTs fusing GNSS fixes with inter-vehicle ranges [23] — we use the worst-pair statistic

$$T_{\mathrm{cons}} = \max_{i<j} \big|\, r_{ij} - 1 \,\big|, \qquad \text{alarm if } T_{\mathrm{cons}} > \lambda. \tag{10}$$

Unlike (8), this test fires on *any* deformation of the formation, in either direction, and is therefore the natural candidate against a distributed attacker.

C. Threshold calibration and metrics

Following the Neyman–Pearson practice of [3], thresholds are calibrated on the H_0 (no-spoofing) scenario, simulated with the same noise levels: λ is the empirical P_{FA}-quantile (lower quantile for the collapse tests, upper for the consistency test) of the statistic over all H_0 epochs, with $P_{\mathrm{FA}} = 10^{-2}$. Performance is reported as (i) the detection probability P_{D}, the fraction of attacked epochs that raise an alarm; (ii) ROC curves obtained by sweeping λ; and (iii) the median relative distance distortion $E = \mathrm{med}_{i<j}\,|\hat{d}_{ij} - d_{ij}|/d_{ij}$, which measures the geometric evidence available to any relative-distance detector. Because the epochs of one mission share clocks, trajectory and satellite geometry, the Monte Carlo unit is the *mission*: 95 % confidence intervals on P_{D} are obtained by a paired bootstrap over missions (500 resamples; H_0 and attack of a mission share the random seed, and the threshold is re-calibrated in every resample).

VI. Simulation Results

A. Setup

Table I summarizes the simulation parameters. For each formation spacing $g \in \{5, 10, 20, 50, 100\}$ m we simulate $N_{\mathrm{MC}} = 20$ independent missions (random start time, clock biases and noise), each of 240 epochs, with and without the attack. The simulator and detection code are available in the companion repository.[1]

[1] https://github.com/silviasch01/GPS_spoofing_distribuito

TABLE I
SIMULATION PARAMETERS.

Parameter	Value
Swarm size m	25 (5×5 grid)
Formation spacing g	5–100 m
Mission duration / epoch	1200 s / $T_s = 5$ s
Satellites	24 (6 planes, 55°, 20 200 km)
Pseudorange noise σ_{pr}	3 m
Ranging noise σ_{rng}	0.1 m
Ground transmitters N_{sp}	24 on circle, $R_{\mathrm{sp}} = 10$ km
Transmitter EIRP	1 mW + 1 dB
Ground-to-UAV channel	Eq. (3), [7]
Spoofer power margin (all links)	13.2–29.0 dB
Target false-alarm rate P_{FA}	10^{-2}
Monte Carlo missions per spacing	20

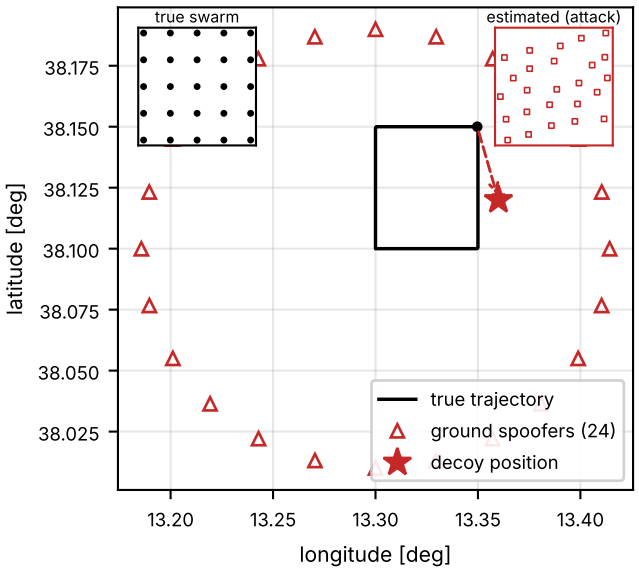


Fig. 1. Attack scenario (one epoch, $g = 20$ m). The swarm (true positions, dots) flies its mission inside a 10 km circle of 24 fixed ground transmitters (triangles), each replaying one satellite. The estimated fixes (open squares) are displaced onto the decoy (star) and mildly deformed.

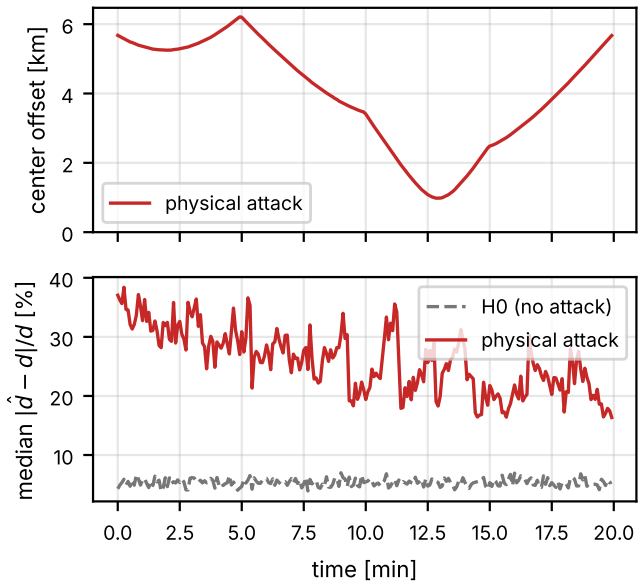


Fig. 2. Per-epoch behaviour at $g = 20$ m (one mission). Top: position offset of the centre UAV. Bottom: median relative-distance error with and without the attack.

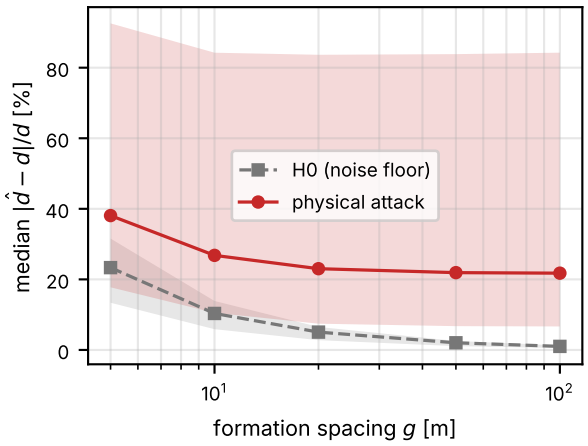


Fig. 3. Median relative-distance distortion vs. formation spacing (median over 20 missions; band: 10th–90th percentile across missions).

B. Attack effectiveness

The attack succeeds: the centre UAV's estimated position is dragged by the full decoy offset, 4.0 km on average over the mission, at every epoch (Fig. 2, top), and every UAV in the swarm is captured on all its visible satellites. The bottom panel shows the price paid by a ground array: the relative-distance distortion is raised above the no-attack noise floor throughout the mission.

C. Formation distortion

Figure 3 reports the median relative distance distortion E over the 20 missions (band: 10th–90th percentile across missions). Under H_0 the distortion is pure GNSS noise and decays as $1/g$ (from 22 % at $g = 5$ m to 0.9 % at 100 m). Under attack it settles at a scale-free plateau of about 22 % of the baseline for $g \geq 20$ m (38 % at 5 m, where noise adds to it): this is the residual of (6), whose magnitude depends on the elevations of the satellites in view and therefore varies between missions (9–26 % at $g = 20$ m). The plateau is above the noise floor for $g \gtrsim 10$ m and below it for compact swarms.

D. Detection performance

Table II and Fig. 4 report P_{D} at $P_{\mathrm{FA}} = 10^{-2}$ with 95 % confidence intervals; Fig. 5 shows the ROC curves at $g = 20$ m. Three observations follow.

First, **the collapse detectors are defeated at every spacing**: all three Jansen variants give $P_{\mathrm{D}} \leq 0.011$ (95 % upper bound), i.e. at or below the false-alarm rate, from 5 to 100 m, and their ROC curves lie on or below the chance diagonal. This is the consequence of Remark 1: the distributed attack translates and

TABLE II
DETECTION PROBABILITY AT $P_{\mathrm{FA}} = 10^{-2}$ AGAINST THE PHYSICAL DISTRIBUTED ATTACK ($\sigma_{\mathrm{pr}} = 3\,\mathrm{m}$, $\sigma_{\mathrm{rng}} = 0.1\,\mathrm{m}$; 20 MISSIONS, 4800 EPOCHS PER CELL; 95 % CI FROM THE PAIRED MISSION BOOTSTRAP).

g [m]	Jansen min	Jansen max	Jansen majority	
5	0.005 [0.004, 0.011]	0.007 [0.004, 0.009]	0.000 [0.000, 0.000]	0.02
10	0.002 [0.000, 0.005]	0.002 [0.001, 0.004]	0.000 [0.000, 0.000]	0.11
20	0.001 [0.000, 0.003]	0.000 [0.000, 0.001]	0.000 [0.000, 0.000]	0.26
50	0.000 [0.000, 0.001]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.56
100	0.003 [0.000, 0.011]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.64

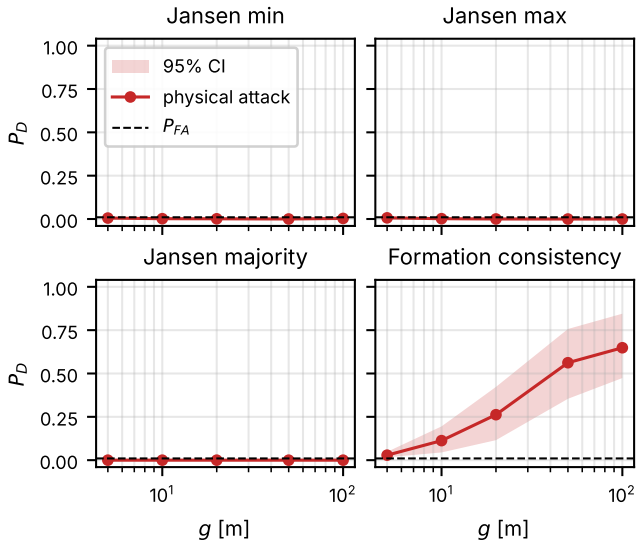


Fig. 4. Detection probability vs. formation spacing at $P_{\mathrm{FA}} = 10^{-2}$ for the four detectors (20 missions; band: 95 % bootstrap CI). The collapse detectors never leave the P_{FA} line.

deforms the formation but never collapses it, so the one-sided statistic of (8) never crosses its threshold; the majority variant, the one recommended in [3], never fires at all. No threshold choice changes this, since the attacked statistic is displaced *away* from the alarm region.

Second, **the consistency test retains partial power that grows with the formation spacing**: P_{D} rises from 0.03 at 5 m and 0.11 at 10 m to 0.56 at 50 m and 0.65 at 100 m, with wide confidence intervals that reflect the mission-to-mission variability of the residual. The test sees the 22 % deformation only where it exceeds the $1/g$ noise floor; for compact swarms, the regime of interest for formation flight, the attack passes essentially undetected at $P_{\mathrm{FA}} = 10^{-2}$, and even at 100 m one attacked epoch in three raises no alarm.

Third, **the residual is the only handle the defender has**. It is not a tuning artifact of the attack (power does not change it) and it is not removable by routing once the array encloses the operating area; it is the geometric cost of keeping the transmitters on the ground, and its magnitude is governed by the satellite elevations through (6).

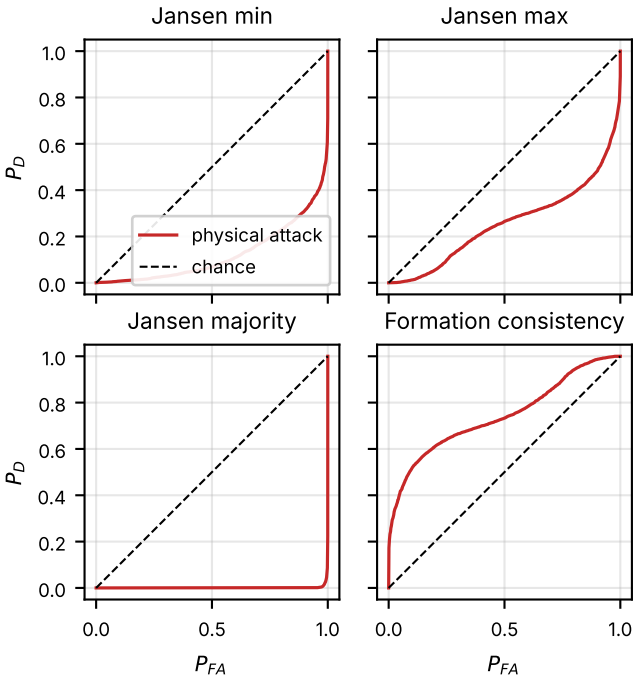


Fig. 5. ROC curves at $g = 20\,\mathrm{m}$ against the physical distributed attack. The collapse detectors operate at or below chance.

VII. DISCUSSION AND LIMITATIONS

Scope of the result: The evaluation covers detectors that are functions of relative position/distance quantities—the collapse family and the two-sided consistency family. The former is blind to the attack for a structural reason (Remark 1); the latter is limited only by the ratio between the ground-array residual and the GNSS noise, and would detect the attack reliably for wide formations, low-noise (e.g. RTK) fixes, or longer integration windows. The result does *not* extend to (i) *direction-of-arrival* detectors: a circle of ground transmitters produces arrivals at near-zero elevation, incongruent with the satellite constellation, so multi-antenna arrays with attitude information [20], [11] would see our attacker, unless its transmitters are placed along the satellite directions as in [17]; nor to (ii) *absolute-reference* checks—ranging to fixed infrastructure, TDoA from ground sensor networks [9], terrain or signal-of-opportunity positioning—for which the 4 km displacement is directly observable; nor to (iii) GNSS/INS detectors such as SemperFi [12] and INS innovation monitors [13], [14], which respond to the time variation of the induced error and were not evaluated here.

Relation to formation-preserving attacks: Tippenhauer et al. [2] prove that a formation-preserving placement exists and GNSS-WASP [17] realizes it with transmitters in line with the satellites at altitude; Park and Yoo [18] show that an exactly common translation is invisible to any relative-only detector. Our attacker is deliberately more constrained—fixed transmitters on the ground—and the quantity we measure is precisely what separates it from those idealized attacks: a

residual deformation of about 22 % of the baseline, enough to be seen by a two-sided test on wide formations and too small to be seen on compact ones. Raising the transmitters would reduce the residual according to (6); we leave that trade-off to future work.

Deployment and steady state: The result assumes an array that encloses the swarm's operating area (Section IV-C); a static array that the swarm leaves deforms the formation by more than 50 % and is readily detected. The decoy position is constant, so the attack is modeled in its steady state after capture; the transient at capture and time-varying decoys are not modeled.

Other simplifications: The simulator uses a spherical Earth, a synthetic constellation, uncorrelated pseudorange noise (spatial correlation would lower the detector's noise floor [3] and favour the consistency test), no multipath or shadowing, and signal capture by power comparison without tracking-loop dynamics.

VIII. Conclusion

We evaluated the classical multi-receiver relative-distance spoofing detectors against a physical distributed ground spoofer attacking a UAV swarm, simulated transmitter by transmitter without any idealization of the attack. The attack displaces the swarm by 4 km and defeats the collapse detectors of Jansen et al. in all their variants at every formation spacing ($P_{\mathrm{D}} \leq 0.011$), because a distributed spoofer does not produce the collapse signature these detectors rely on. A two-sided consistency check against independent ranging retains partial power only for wide formations ($P_{\mathrm{D}} = 0.65$ at 100 m, 0.03 at 5 m), limited by a residual deformation of about 22 % of the baseline that is the geometric cost of ground-based transmitters. Robust swarm navigation therefore cannot rely on collapse detection, and relative consistency checks should be complemented by direction-of-arrival or absolute-reference information.

Acknowledgments

[TODO: ringraziamenti, funding]

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